

# 5.13.82

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**Question :** Let  $\alpha$  and  $\beta$  be the distinct roots of the equation  $x^2 + x - 1 = 0$ . Consider the set  $T = \{1, \alpha, \beta\}$ . For a  $3 \times 3$  matrix  $\mathbf{M} = (a_{ij})_{3 \times 3}$ , define  $R_i = a_{i1} + a_{i2} + a_{i3}$ ,  $C_j = a_{1j} + a_{2j} + a_{3j}$ , for  $i, j = 1, 2, 3$ . Match the following:

- (A) The number of matrices  $\mathbf{M} = (a_{ij})_{3 \times 3}$  with all entries in  $T$  such that  $R_i = C_j = 0$  for all  $i, j$ , is
- (B) The number of symmetric matrices  $\mathbf{M} = (a_{ij})_{3 \times 3}$  with all entries in  $T$  such that  $C_j = 0$  for all  $j$ , is
- (C) Let  $\mathbf{M} = (a_{ij})_{3 \times 3}$  be a matrix with all entries in  $T$  such that  $R_i = 0$  for all  $i$ . Then the absolute value of the determinant of  $\mathbf{M}$  is

- |       |             |      |
|-------|-------------|------|
| 1) 1  | 3) infinite | 5) 0 |
| 2) 12 | 4) 6        |      |

The correct option is:

- |                                                        |                                                        |
|--------------------------------------------------------|--------------------------------------------------------|
| 1) $A \rightarrow 4, B \rightarrow 2, C \rightarrow 1$ | 3) $A \rightarrow 2, B \rightarrow 4, C \rightarrow 3$ |
| 2) $A \rightarrow 2, B \rightarrow 4, C \rightarrow 5$ | 4) $A \rightarrow 1, B \rightarrow 5, C \rightarrow 4$ |

**Solution :**

Let  $T = \{1, \alpha, \beta\}$  where  $\alpha, \beta$  are distinct roots of  $x^2 + x - 1 = 0$ . For a  $3 \times 3$  matrix  $M = (a_{ij})$ , define

$$R_i = a_{i1} + a_{i2} + a_{i3}, \quad C_j = a_{1j} + a_{2j} + a_{3j}. \quad (4.1)$$

$$\alpha + \beta = \frac{-\text{coefficient of } x}{\text{coefficient of } x^2} = -1 \quad (4.2)$$

$$1 + \alpha + \beta = 0 \quad (4.3)$$

(A) We need to find matrices  $\mathbf{M}$  with all entries in  $T$  such that  $R_i = C_j = 0$  for all  $i, j$ , So

$$R_1 = R_2 = R_3 = C_1 = C_2 = C_3 = 0. \quad (4.4)$$

Only possible case is some permutation of  $\{1, \alpha, \beta\}$  as  $1 + \alpha + \beta = 0$ .

As both row total = column total = 0.

**Latin Square :**

Latin Square is a Matrix in which one element occurs exactly once in a row and a column.

The Matrix  $\mathbf{M}$  should be Latin Square of order 3 on the symbol set  $\{1, \alpha, \beta\}$

By permutations , we get

$$3 \times 2 \times 1 \times 2 = 12 \quad (4.5)$$

One of the 12 Latin Square Matrices is given as

$$\mathbf{M} = \begin{pmatrix} 1 & \alpha & \beta \\ \alpha & \beta & 1 \\ \beta & 1 & \alpha \end{pmatrix} \quad (4.6)$$

The number of Matrices  $\mathbf{M} = 12$ .

Thus,  $A = 2$ .

(B) Symmetric matrices  $\mathbf{M}$  with entries in  $T$  such that  $C_j = 0$ .

$$\mathbf{M} = \mathbf{M}^T \quad (4.7)$$

So both  $R_i = C_j = 0$  for all  $i, j$ .

**Latin Square :**

So The Matrix  $\mathbf{M}$  should be Latin Square of order 3 on the symbol set  $\{1, \alpha, \beta\}$  and satisfying  $\mathbf{M} = \mathbf{M}^T$ .

By permutations, we get

$$3 \times 2 \times 1 = 6 \quad (4.8)$$

One of the 6 Latin Square Matrices is given as

$$\mathbf{M} = \begin{pmatrix} 1 & \alpha & \beta \\ \alpha & \beta & 1 \\ \beta & 1 & \alpha \end{pmatrix} \quad (4.9)$$

The number of Matrices  $\mathbf{M} = 6$ .

Thus,  $B = 4$ .

(C) Matrices  $\mathbf{M}$  with entries in  $T$  such that  $R_i = 0$  for all  $i$ . So each row is a permutation of  $(1, \alpha, \beta)$ .

$$1 + \alpha + \beta = 0 \quad (4.10)$$

If rows are identical

$$|\mathbf{M}| = 0 \quad (4.11)$$

If rows are distinct then For matrix  $\mathbf{M}$  if  $C_1 \rightarrow C_1 + C_2 + C_3$  then every element of  $C_1$  is 0.

So  $|\mathbf{M}| = 0$ .

Thus  $C = 5$ .

$$\boxed{A = 2, B = 4, C = 5} \quad (4.12)$$

Correct option is (b).